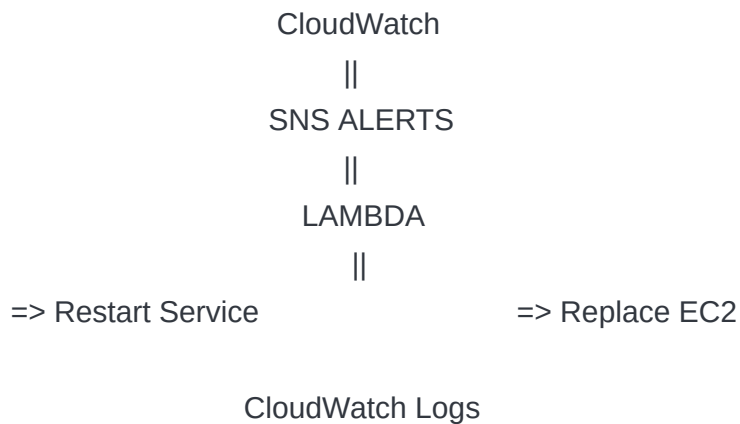


Project

PROJECT: Cloudwatch watches server, when something goes wrong, Lambda fixes it automatically without need for human intervention.

- working:
1. Detect – CloudWatch tracks CPU, memory, and disk in real time
 2. Alert – if a threshold is crossed, an alarm fires and notifies SNS
 3. Trigger – SNS invokes a Lambda function
 4. Remediate – Lambda restarts the service (soft fix) or replaces the EC2 instance (hard fix)
 5. Log – every action is recorded in CloudWatch Logs as an incident
 6. Result – system heals itself, with a full audit trail, without anyone SSH-ing in

Flow chart:

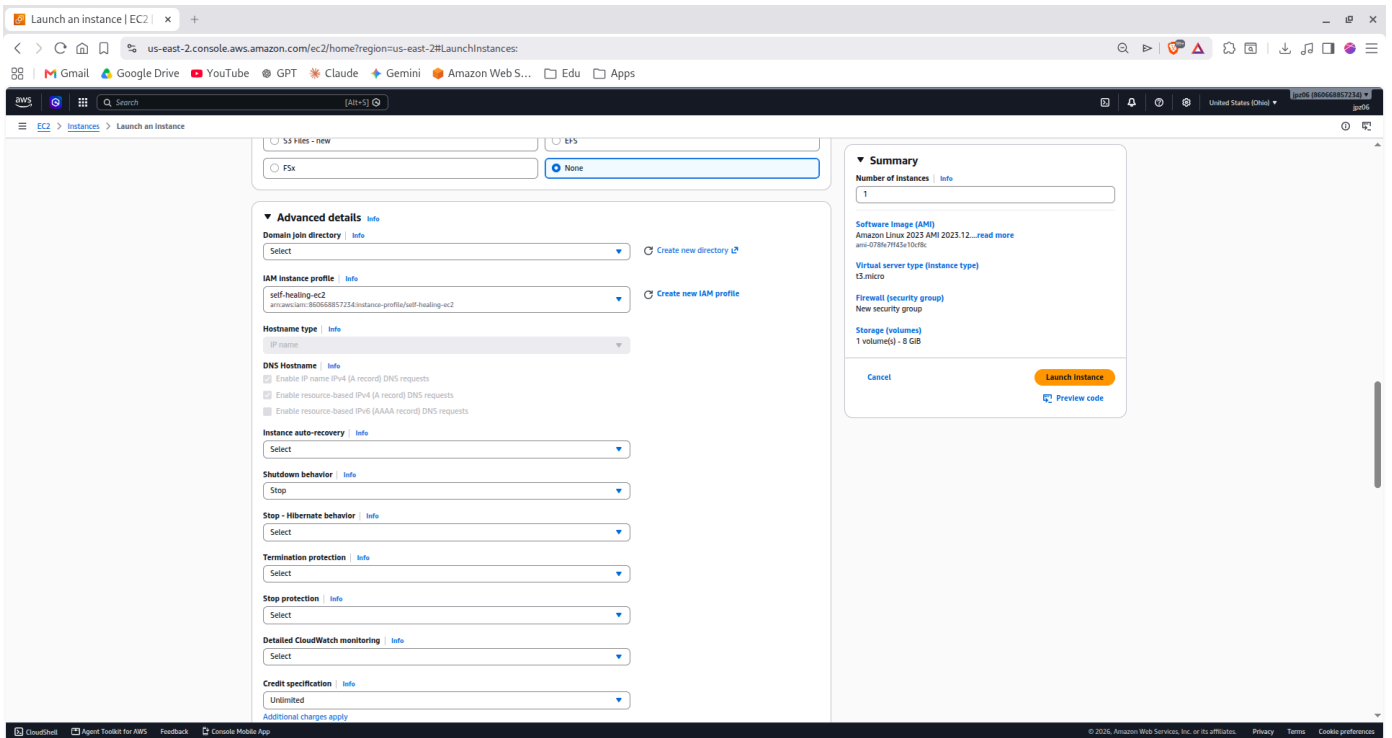


**IAM Role

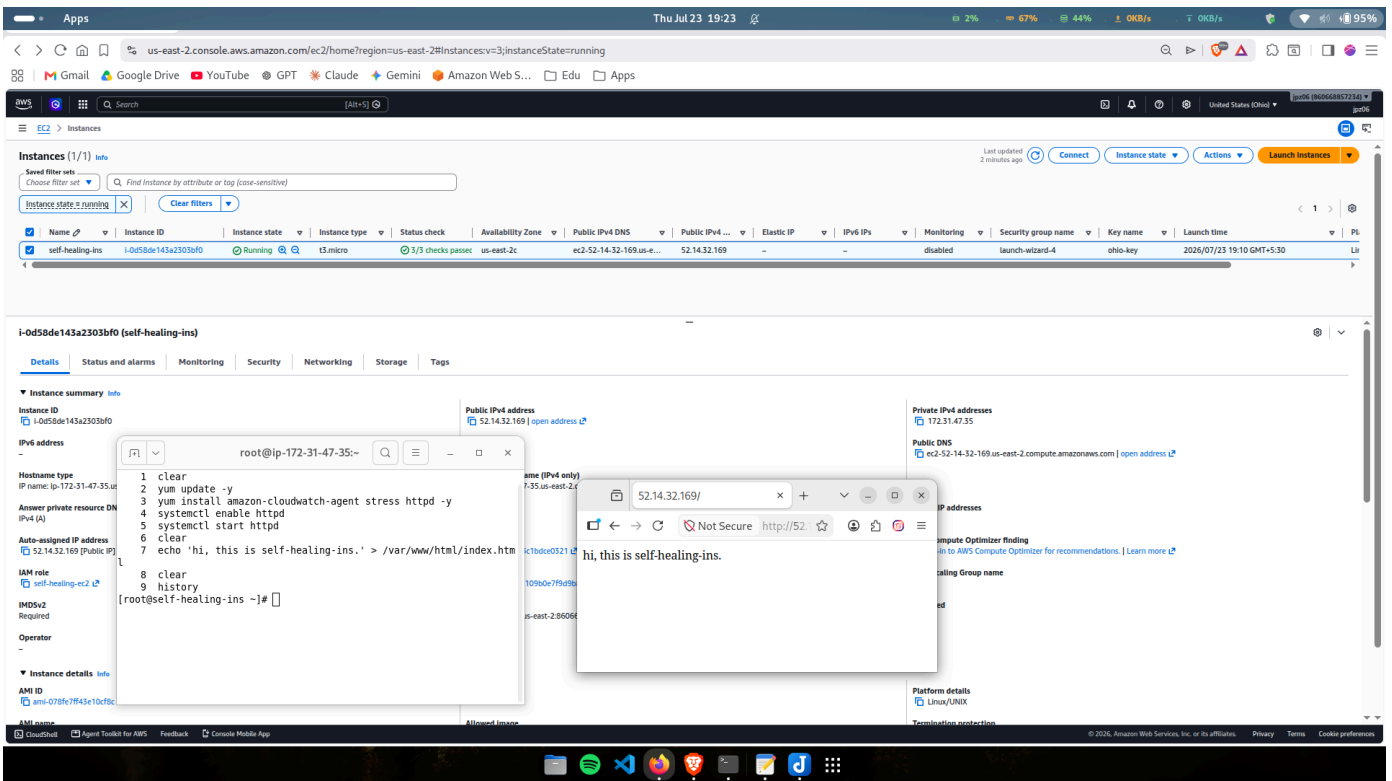
###STEPS & SS.

- a. EC2 Instance Role ('self-healing-ec2' role)

**



installed amazon cloudwatch agents, httpd
on pasting public-ip, result : index.html shows up.



===>>> SNS-created

Instances | EC2 | us-east-2

Subscription: 3d5d738c-2a3f-46a2-9080-c40d0a4c9b1f

us-east-2.console.aws.amazon.com/sns/v3/home?region=us-east-2#/subscription/arn:aws:sns:us-east-2:860668857234:self-healing-alerts:3d5d738c-2a3f-46a2-9080-c40d0a4c9b1f

Gmail Google Drive YouTube GPT Claude Gemini Amazon Web S... Edu Apps

Amazon SNS

Dashboard
Topics
Subscriptions

▼ Mobile
Push notifications
Text messaging (SMS)

Subscription: 3d5d738c-2a3f-46a2-9080-c40d0a4c9b1f

Edit Delete

Details
ARN
arn:aws:sns:us-east-2:860668857234:self-healing-alerts:3d5d738c-2a3f-46a2-9080-c40d0a4c9b1f
Endpoint
jonathanh7353@gmail.com
Topic
[self-healing-alerts](#)

Status
Confirmed
Protocol
EMAIL

Subscription filter policy Redrive policy (dead-letter queue)

Subscription filter policy [Info](#)
This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.
[Edit](#)

a. High CPU Alarm

Instances | EC2 | us-east-2

Create alarm | Alarms | CloudWatch

us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2/create?~(Page~Preview~metricType~classic~AlarmType~MetricAlarm~AlarmData~(AlarmName~highCPU-self-healing~AlarmDescr...

Gmail Google Drive YouTube GPT Claude Gemini Amazon Web S... Edu Apps

CloudWatch

Alarms

Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1: Specify metric and conditions

Step 2: Configure actions

Step 3: Add alarm details

Step 4: [Preview and create](#)

Preview and create

Step 1: Specify metric and conditions

Metric
Graph
This alarm will trigger when the blue line goes above the red line for 2 datapoints within 2 minutes.

Percent
70

Namespaces
AWS/EC2

Metric name
CPUUtilization

InstanceId
i-0d58de143a2303bf0

Instance name
self-healing-ins

Statistic
Average

Period
1 minute

Conditions
Threshold type
Static
Whenever CPUUtilization is Greater (>) than... 70

Additional configuration
Datapoints to alarm
2 out of 2
Missing data treatment
Treat missing data as missing

B **.high memory alarm**

Create alarm

Metrics
Monitor CloudWatch metrics

Type
Choose the metric type for this alarm.
☒ Classic
Use the standard CloudWatch metric selection experience.

Metric
This alarm will trigger when the blue line goes above the red line for 2 datapoints within 2 minutes.

Graph
Percent
80%
54.2%
28.4%
12:15 12:30 12:45 13:00 13:15 13:30 13:45 14:00 14:15 14:30 14:45 15:00
mem_used_percent

Namespaces
SelfHealingDemo

Metric name
mem_used_percent

host
self-healing-ins

Statistic
Average

Period
1 minute

Conditions
Threshold type
☒ Static
Use a value as a threshold

Whenever mem_used_percent is...
Define the alarm condition.
☒ Greater
Greater/Equal
Lower/Equal
Lower

c. high disk alarm

Successfully created alarm high-disk-alarm.

Alarms (3)

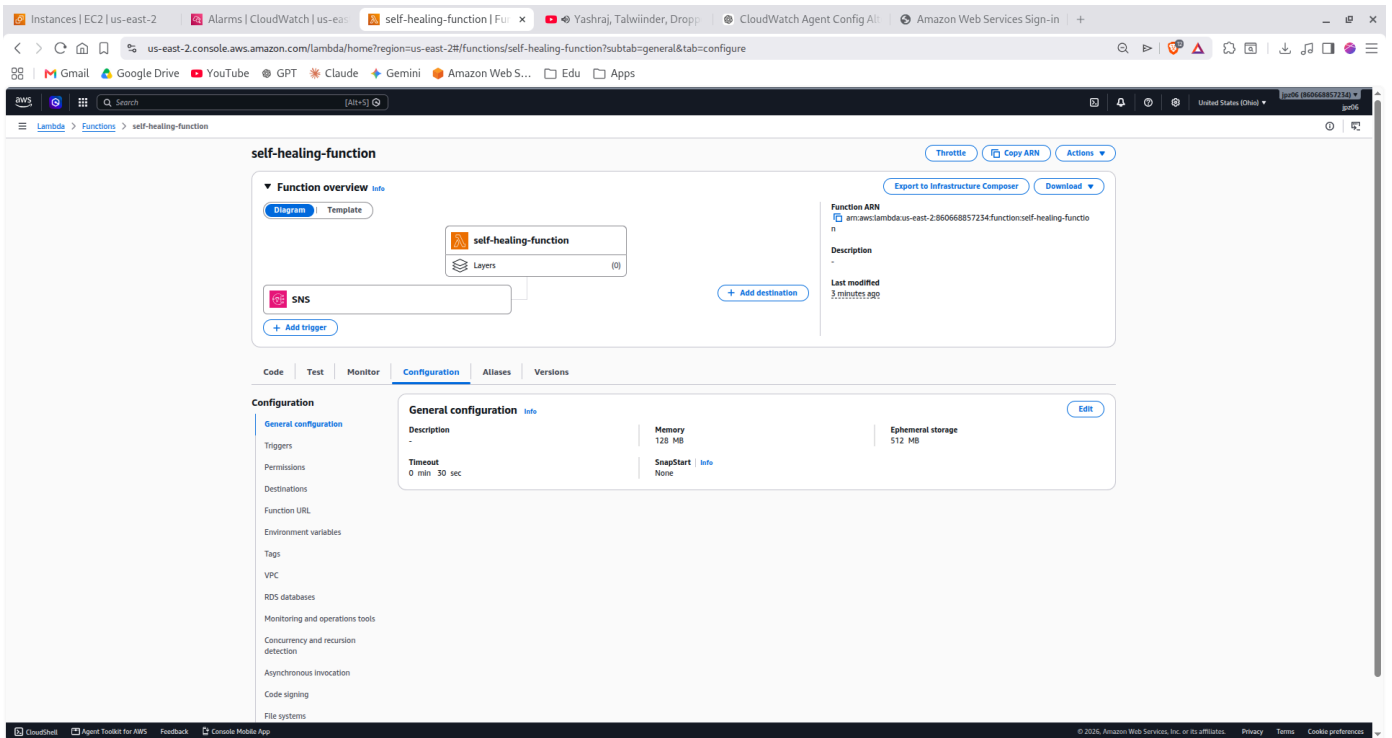
Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
high-disk-alarm	Insufficient data	Actions enabled	-	2026-07-23 15:08:36	disk_used_percent > 85 for 2 datapoints within 2 minutes
high-CPU-self-healing	OK	Actions enabled	-	2026-07-23 15:07:20	CPUUtilization > 70 for 2 datapoints within 2 minutes
high-memory-alarm	OK	Actions enabled	-	2026-07-23 15:07:03	mem_used_percent > 80 for 2 datapoints within 2 minutes

==>> Lambda

==>> deployed python code(source : chat-gpt)

==>> SNS Trigger added.

==>> timeout set to 30s

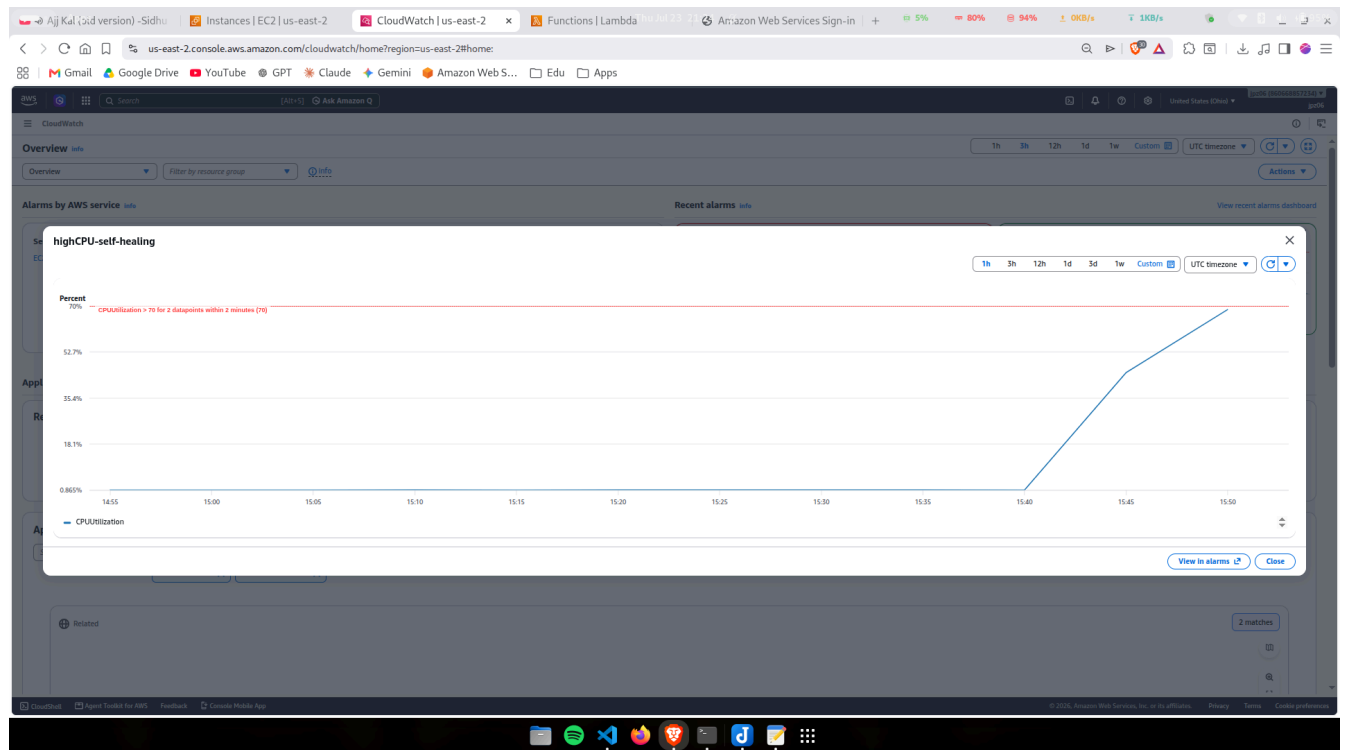


• **TEST**

--- trigger cpu alarm

[command used: stress --cpu 2 --timeout 300]

cloudwatch monitoring:



****logs****

The screenshot shows the AWS CloudWatch console. The left sidebar has a menu with 'Log Management' selected. The main area displays 'Log events' for the function 'aws/lambda/self-healing-function'. A search bar is at the top. Below it, a table of log events is shown. The first event is a 'START' event with a message 'INIT_START Runtime Version: python:3.14.mainline.v59 Runtime Version ARN: arn:aws:lambda:us-east-2::runtime:91682077d7f37ed39c054f7b4d896a70670e4995535d336206df5d249990a52'. The second event is a 'START' event with a message 'START RequestId: 6d67c064-b40f-4675-9ab7-8576edf557e4 Version: SLATEST'. The third event is an 'END' event with a message 'END RequestId: 6d67c064-b40f-4675-9ab7-8576edf557e4'. The fourth event is a 'REPORT' event with a message 'REPORT RequestId: 6d67c064-b40f-4675-9ab7-8576edf557e4 Duration: 1875.61 ms Billed Duration: 1718 ms Memory Size: 128 MB Max Memory Used: 183 MB Init Duration: 642.22 ms'. The table has columns for 'Timestamp' and 'Message'.

AUTO SCALING GROUP + LOAD BALANCER

The screenshot shows the AWS console for Auto Scaling groups and Load balancers. At the top, a green banner states: 'self-heal-autoscaling, 1 Load balancer, 1 Target group, 1 Listener created successfully. 1 new target group has been attached to ASG.' Below this, the 'Auto Scaling groups (1)' section is shown. It has a search bar and a table with columns: Name, Status, Launch template/configuration, Instances, Instance health, Desired capacity, Min, Max, Availability Zones, and Creation time. The table shows one group, 'self-heal-autoscaling', with status 'Updating capacity', launch template 'self-heal-template', version 'Default', 0 instances, and 1 desired capacity. Below this, the 'Load balancers (1)' section is shown. It has a search bar and a table with columns: Name, State, Type, Scheme, IP address type, VPC ID, Availability Zones, Security groups, DNS name, and ARN. The table shows one load balancer, 'self-heal-loadbalancer', with state 'Active', type 'application', scheme 'Internet-facing', IP address type 'IPv4', VPC ID 'vpc-0d4a493c1bdce0321', 3 availability zones, security group 'sg-0f6e70d225adaf043.4', and DNS name 'self-heal-loadbalancer-110...'. The table has a 'Create Auto Scaling group' button and a 'Create load balancer' button.

1. closes the ongoing instance and starts a new instance.

The screenshot shows a terminal window with the prompt 'jp@lnvo: ~/Downloads/keys'. The terminal output shows a connection error: '[root@self-healing-ins ~]# Connection to 52.14.32.169 closed by remote host. Connection to 52.14.32.169 closed. jp@lnvo:~/Downloads/keys\$'.

2.new instance is created,

Currently creating AMI ami-0c846768ce29aacfc from Instance i-0d58de143a2303bf0. Check that the AMI status is 'Available' before deleting the instance or carrying out other actions related to this AMI.

Instances (2) Info

Save filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring
☐		i-0b0348791f690ca9e	Running	t3.micro	3/3 checks passed	us-east-2a	ec2-18-217-144-116.us...	18.217.144.116	-	-	disabled
☐	self-healing-ins	i-0d58de143a2303bf0	Running	t3.micro	3/3 checks passed	us-east-2c	ec2-52-14-32-169.us-e...	52.14.32.169	-	-	disabled

All instances

Route 53

Route 53 > Hosted zones > jonathanpopat.com

Record for jonathanpopat.com was successfully created.
Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.

Public jonathanpopat.com Info

Delete zone Test record Configure query logging

Hosted zone details

Edit hosted zone

Records (3) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (3) Info

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value

	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalu...	Reco...
☐	jonathanpopat.com	A	Simple	-	Yes	self-heal-loadbalancer-1104...	-	-	Yes	-
☐	jonathanpopat.com	NS	Simple	-	No	ns-227.awsdns-28.com. ns-1667.awsdns-16.co.uk. ns-1093.awsdns-08.org. ns-902.awsdns-48.net.	172,800	-	-	-
☐	jonathanpopat.com	SOA	Simple	-	No	ns-227.awsdns-28.com. awsd...	900	-	-	-

1. created an hosted zone
2. targeted to a load balancer by A-record, using simple routing